NOTE: The document identifier and heading have been changed on this page to reflect that this is a performance specification. There are no other changes to this document. The document identifier on subsequent pages has not been changed, but will be changed the next time this document is revised.

MIL-PRF-15733/51E
10 September 1986
SUPERSEDING
MIL-F-15733/51D
23 September 1982

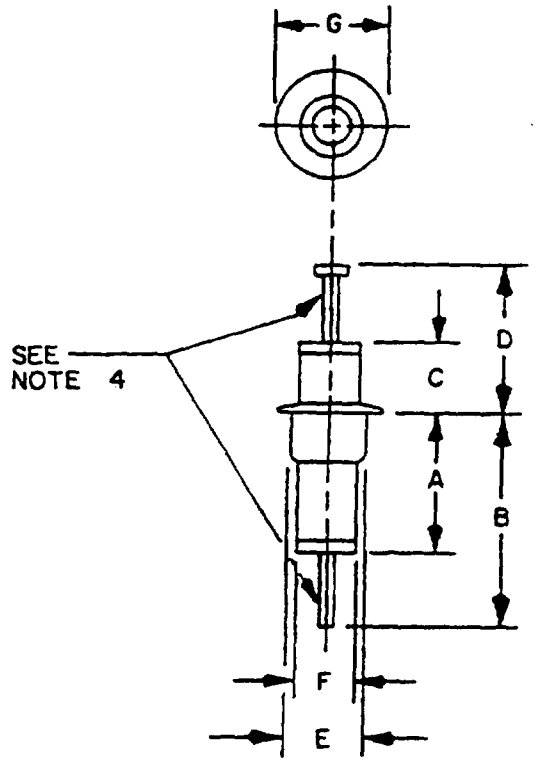
PERFORMANCE SPECIFICATION SHEET

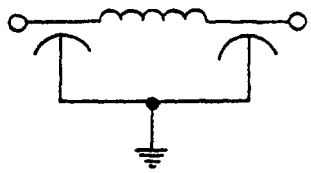
FILTERS, RADIO FREQUENCY INTERFERENCE,

STYLE FL48

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the filters described herein shall consist of this specification and the latest issue of MIL-F-15733.





TYPICAL CIRCUIT DIAGRAM

NOTES:
1. Dimensions (see table I).
2. Circuit diagram is for information only.
3. Suggested mounting hole diameter: .160 (4.06 mm).
4. Leads shall be solid, tinned, AWG 20, .032 (0.81mm) diameter.

FIGURE 1. Dimensions and configuration

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TABLE I. Filter dimensions.

Dash number	A		B		C		D		E		F		G	
	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
0001	.219 (5.56)	.281 (7.14)	.375 (9.52)	.437 (11.10)	.125 (3.18)	.187 (4.75)	.281 (7.14)	.343 (8.71)	.135 (3.43)	.150 (3.81)	.105 (2.67)	.140 (3.56)	.185 (4.70)	.205 (5.21)
0002	.204 (5.18)	.249 (5.94)	.749 (19.02)	.811 (20.60)	.141 (3.58)	.186 (4.34)	.281 (7.14)	.343 (8.71)	.130 (3.30)	.160 (4.06)	NA	NA	.188 (4.78)	.218 (5.54)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only.

TABLE II. Electrical characteristics.

Dash number	Operating temperature range	Rated voltage (volts)	Weight max (grams)	Capacitance min (pF)	1/ Minimum insertion loss (db) in accordance with MIL-STD-220, at										
					+25°C						-55°C and maximum operating temperature				
					5 MHz	10 MHz	20 MHz	100 MHz	1 GHz	10 GHz	5 MHz	10 MHz	20 MHz	100 MHz	10 GH
0001	-55°C to +85°C	200 V dc 200 V rms	.8	5,500	---	15	21	55	70	70	---	3	8	40	70
0002	-55°C to +125°C	70 V dc	.8	22,000	20	25	20	70	70	70	5	10	5	65	70

1/ Insertion loss measurements shall be performed at full-load from 5 MHz to 20 MHz and at no-load above 20 MHz.

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REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Case: Not applicable (metallic mounting eyelet).

Terminals: Solderable.

Operating temperature range: See table II.

Rated voltage: See table II.

Rated current: 10 amperes, dc or ac (rms).

Insertion loss: See table II.

Seal: Not applicable.

Capacitance to ground: Method 305 of MIL-STD-202. Test frequency: 1,000 ±100 Hz. Measured capacitance shall be in accordance with table II.

Temperature rise: 25°C, maximum.

Insulation resistance: Method 302, test condition A of MIL-STD-202. Insulation resistance measured at 25°C between either terminal and the case shall be at least 20,000 megohms.

Test potential: Rated dc voltage.

Voltage drop: Not applicable.

Insertion loss: In accordance with MIL-F-15733 at 25°C and at temperature extremes.

Overload: In accordance with MIL-F-15733.

Measurements at +25°C after test:

Insulation resistance shall be measured and shall meet initial requirements.

Terminal strength: Method 211, test condition A of MIL-STD-202.

Applied force: 5 pounds.

Flashpoint of impregnant or potting compound: Not applicable.

Salt spray (corrosion): Not applicable.

Thermal shock: Not applicable.

Immersion: Not applicable.

Shock (specified pulse): Method 213, test condition I of MIL-STD-202.

Electrical load: 100 percent of the rated voltage between terminals and case.

Measurement after test: Insertion loss (check test) shall meet initial requirements.

Vibration, high frequency: Method 204, test condition D of MIL-STD-202.

Electrical load: Rated voltage and rated current shall be applied continuously during test.

Test during vibration: During the last cycle in each direction, an electrical measurement shall be made to determine intermittent-open or short-circuiting.

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Moisture resistance: Method 107, test condition A of MIL-STD-202; the temperature of step 3 shall be +85°C, +3°C, -0°C (dash number 0001) or +125°C, +3°C, -0°C (dash number 0002). No measurement shall be made before and after cycling. Filters shall then be tested in accordance with method 106 of MIL-STD-202.

Initial measurements: Not applicable.

Polarization voltage: Not applicable.

Loading voltage: Not applicable.

Measurements after 24-hour drying period at a temperature not to exceed +85°C (dash number 0001) or +125°C (dash number 0002), and a relative humidity of 50 percent:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 1 gigohm.

Insertion loss (check test) shall meet the initial requirements.

Life: Method 108, test condition D of MIL-STD-202.

Test temperature:

Dash number 0001: +85°C.
Dash number 0002: +125°C.

Test voltage: 1.2 times the rated rms voltage.

Test current: 10 amperes.

Measurements after test:

Dielectric withstanding voltage with 90 percent of the initial voltage applied for 5 ±1 seconds.

Insulation resistance shall be not less than 5,000 megohms.

Insertion loss (check test) shall meet the initial requirements.

Solderability of mounting termination: In accordance with MIL-F-15733.

Marking: Filters shall not be marked. Full marking, in accordance with MIL-F-15733, shall be marked on the unit package.

Part number: M15733/51-(dash number from table II).

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Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - AR, MI, SL
Navy - SH
Air Force - 11,17, 99
DLA - ES

User activities:
Army - AT, AV, ME,
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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